TokTickIT - Sprint 4 Engineering Specification

Course: CPE 334, Semester 1/2026
Sprint: Lab 4 Actions Taken, Dashboards, and Workflow Hardening
Depends on: Lab 3 `specification.md`, `ui-spec.md`, `api-spec.md`, `tests.md` (all merged to `main`)

1. Sprint Goal

Sprint 4 completes the TokTickIT service-desk workflow. IT Staff and Administrators can now record **Actions Taken** against a Ticket, the Ticket status lifecycle is fully enforced end to end (including a backend-enforced resolution gate), and Requesters, IT Staff, and Administrators each get a concise, role-appropriate **Dashboard** that links back to the detailed screens in scope. The sprint closes with accessibility/visual hardening and safe-failure checks for the Lab 4 workflows.

2. Stakeholder Request (interpreted)

The stakeholder wants a durable, auditable record of the actual work IT Staff perform on a Ticket — not just comments and notes, but discrete actions with a date/time, description, outcome, and an explicit follow-up flag. A Requester's own claim that "this looks fixed" should never be enough to close a Ticket by itself; an IT Staff member must review the work and formally move the Ticket through its status lifecycle. Both Requesters and IT Staff should be able to see, at a glance, what needs their attention next, without replacing the detailed Ticket screens they already use.

3. Scope

3.1 Included

- Actions Taken: create, list, and edit, scoped to a Ticket.
- The complete Ticket status-transition matrix, including a backend-enforced Resolved gate.
- Requester Dashboard and IT Staff Dashboard (Administrator reuses the IT Staff Dashboard).
- Database/Prisma increments, migration, and idempotent seed data for the above.

- REST API increments for Actions Taken, status transitions, and dashboards.
- Zen Green UI extensions for all new screens, responsive and accessible.
- Final hardening of the Lab 4 workflows (safe failure, duplicate-submit prevention, console/UI cleanup).

3.2 Explicitly excluded (per handout §4.2)

- Automatic SLA clocks, escalation engines, on-call scheduling, breach notifications.
- Email, SMS, LINE, push, or other external notifications.
- Inventory/spare-parts management, purchasing, or cost accounting.
- Time-sheet billing, payroll, or labor-cost calculation.
- Multi-level approval workflows or electronic signatures.
- Advanced BI tools, custom report builders, export warehouses.
- Multi-tenant organizations or production-scale cloud operations.
- Any feature not covered by this contract.

4. Functional Requirements

ID	Requirement
FR-01	IT Staff and Administrator can create an Action Taken on any Ticket they are authorized to access.
FR-02	IT Staff, Administrator, and the owning Requester can view the full list of Action Taken records for a Ticket, ordered by Action Date/Time (most recent first).
FR-03	IT Staff and Administrator can edit an existing Action Taken's Description, Result, Follow-Up Required, Follow-up Note, and Attachment Notes.
FR-04	Requesters see Actions Taken read-only; no create/edit controls are shown or accepted from a Requester session.
FR-05	Performed By is set automatically from the authenticated session on create and is never accepted from client input.
FR-06	IT Staff/Administrator can transition a Ticket's status from Ticket Detail, restricted to the matrix in §5.2.
FR-07	The backend enforces the Resolved-transition gate (BR-07) independent of what the UI shows.

ID	Requirement
FR-08	Requesters can continue to flag a Ticket as "appears resolved" (carried from Lab 3); this remains advisory only.
FR-09	The system provides a Requester Dashboard summarizing only the authenticated Requester's own Tickets.
FR-10	The system provides an IT Staff Dashboard summarizing operational data across the Tickets that user is authorized to see.
FR-11	The system provides an Administrator Dashboard that reuses the IT Staff Dashboard and adds concise user-account counts.
FR-12	Every dashboard card/list item supports drill-down navigation to the corresponding filtered Ticket Queue, My Tickets, or Ticket Detail view.
FR-13	Loading, empty, forbidden, conflict, not-found, and safe-failure states are consistent across the Lab 4 screens and workflows.
FR-14	Duplicate submissions (double-click, network retry) on Action Taken creation and status transitions are prevented or safely deduplicated.

5. Business Rules

5.1 Actions Taken

ID	Rule
BR-01	An Action Taken belongs to exactly one Ticket.
BR-02	The Ticket Owner coordinates the Ticket as a whole, but an Action Taken may be recorded by a different IT Staff member.
BR-03	Performed By is always the authenticated actor from the session; the client cannot set or override it.
BR-04	Follow-up Note is required (non-empty, trimmed) when Follow-Up Required is `true` ; it is cleared/ignored when `false` .

ID	Rule
BR-05	Action Description is required, trimmed, and at least 3 characters.
BR-06	Action Date/Time must be on or after the Ticket's creation date/time and not later than the current server time.
BR-15	Any IT Staff/Administrator with access to the Ticket may edit any Action Taken on it, not only the one who originally recorded it (consistent with BR-02).
BR-16	Action Taken records can only be created while the Ticket's status is not `Closed` or `Cancelled`; the Ticket must first be Reopened (BR-10).

5.2 Ticket Status and Resolution

Statuses (unchanged from the handout): `New`, `Open`, `In Progress`, `Waiting for Requester`, `Resolved`, `Closed`, `Reopened`, `Cancelled`.

ID	Rule
BR-07	A Ticket transitions to `Resolved` only when the request comes from IT Staff/Administrator **and** the Ticket already has at least one Action Taken record with Result = `Resolved`, recorded by IT Staff/Administrator.
BR-08	A Requester's "appears resolved" flag (Lab 3) is stored as advisory only; it never triggers the `Resolved` transition itself — BR-07 always governs.
BR-09	The backend rejects any status transition not present in the matrix below with `409 Conflict`, regardless of what the client UI currently shows.
BR-10	Only Administrators may move a Ticket out of `Cancelled` or `Closed` (override reopen); IT Staff cannot.
BR-11	Updates to a Ticket or an Action Taken use optimistic concurrency: the client must send the record's last-seen `updatedAt`; a mismatch is rejected with `409 Conflict` and the current record is returned so the client can refetch.

Ticket status-transition matrix

From	To	Allowed roles	Precondition
New	Open	IT Staff, Administrator	—
New	Cancelled	Requester, IT Staff, Administrator	—
Open	In Progress	IT Staff, Administrator	Ticket must have an Owner (auto-assigned to the acting user if unset)
Open	Waiting for Requester	IT Staff, Administrator	—
Open	Cancelled	IT Staff, Administrator	—
In Progress	Waiting for Requester	IT Staff, Administrator	—
In Progress	Resolved	IT Staff, Administrator	BR-07 gate
In Progress	Cancelled	IT Staff, Administrator	—
Waiting for Requester	In Progress	IT Staff, Administrator	—
Waiting for Requester	Cancelled	IT Staff, Administrator	—
Resolved	Closed	IT Staff, Administrator	—
Resolved	Reopened	Requester, IT Staff, Administrator	—
Closed	Reopened	Administrator only	BR-10
Reopened	In Progress	IT Staff, Administrator	—
Reopened	Cancelled	IT Staff, Administrator	—

From	To	Allowed roles	Precondition
Cancelled	Reopened	Administrator only	BR-10

5.3 Dashboards

ID	Rule
BR-12	Every dashboard metric is computed by the backend from live Ticket/Action Taken data at request time; the client never derives a count locally.
BR-13	IT Staff dashboard: `Unassigned` = Tickets with no Owner and status not in `{Closed, Cancelled}`. `My Assigned` = Tickets where Owner = current user and status not in `{Closed, Cancelled}`.
BR-14	The Requester Dashboard is always scoped server-side to Tickets where Requester = current session user, regardless of any client-supplied filter or id.

6. UI Specification Summary

Full detail lives in `ui-spec.md`. Summary:

- **IT Staff Dashboard** (`/dashboard` , staff/admin view) — six metric cards (Unassigned, My Assigned, New, Open, In Progress, Waiting for Requester), a by-IT-Priority breakdown, a Recent Tickets list, and Quick Actions. Every card and row is a drill-down link.
- **Requester Dashboard** (`/dashboard` , requester view) — four metric cards (Total Open, Waiting for You, Recently Resolved, Closed), a Recent Tickets list, and Quick Actions.
- **Actions Taken** — a new, filled-in "Service Actions" tab on the existing Ticket Detail tab bar (previously a Lab 3 placeholder), with a list/table view and an accessible create/edit form.
- **Ticket Workflow controls** — a status control on Ticket Detail that only offers permitted next statuses, with the backend as final authority and a plain-language reason shown when the Resolved gate isn't met yet.
- All new screens reuse the existing Zen Green tokens, card/table/badge/button/form components, with consistent loading/empty/forbidden/error states.

7. Data Changes

7.1 New tables

`action_taken`

Field	Type	Notes
`id`	integer, PK	Matches existing integer-ID convention (not UUID).
`ticket_id`	integer, FK → `ticket.id`	BR-01. Indexed.
`action_at`	timestamp	Client-supplied, bounded by BR-06.
`description`	text	BR-05.
`result_id`	integer, FK → `action_result.id`	See §7.2.
`performed_by_user_id`	integer, FK → `user.id`	Server-set only (BR-03). Indexed.
`follow_up_required`	boolean	Default `false`.
`follow_up_note`	text, nullable	Required when `follow_up_required = true` (BR-04).
`attachment_notes`	text, nullable	Free text, e.g. "see screenshot_2026.png".
`created_at`	timestamp	Audit; server-set.
`updated_at`	timestamp	Audit + optimistic-concurrency token (BR-11).

`action_result` (reference table, same pattern as existing `status` / `priority` / `category`)

Seed values: `In Progress`, `Resolved`, `Escalated`, `No Fault Found`, `Awaiting Parts / Access`, `Duplicate / No Action Needed`. The literal value `Resolved` is what BR-07's gate checks against.

7.2 Indexes

- `action_taken(ticket_id)`, `action_taken(performed_by_user_id)`
- `ticket(status_id)`, `ticket(owner_user_id)` (support dashboard aggregate queries)

7.3 Migration and backfill

The migration is additive only and does not alter existing records. Existing Tickets get zero `action_taken` rows; the Actions Taken UI and all dashboard counts must treat "zero rows" as a normal, defined empty state rather than an error (AC-11). Rollback drops the two new tables and their FKs.

7.4 Seed data

Idempotent (safe to re-run): upserts `action_result` reference rows by name, and seeds a representative spread of Tickets — some with zero, one, and multiple Action Taken rows, across every status, with both assigned and unassigned ownership — so every dashboard metric has at least one non-zero example and at least one zero/empty example to demonstrate.

7.5 Design decisions (justification)

1. **`action_result` as a reference table, not free text or a Prisma enum** — consistent with how `status` / `priority` / `category` are already modeled in this codebase, and it lets BR-07's gate do an exact, typo-proof match on `Resolved` instead of fuzzy string comparison.
2. **Optimistic concurrency via `updated_at` rather than a new integer `version` column** — reuses a field every record already has, keeps the same conflict-detection approach on both `ticket` and `action_taken`, and needs no extra schema surface.

8. API Contract

Full detail lives in `api-spec.md`. New endpoints:

- `POST /api/tickets/:ticketId/actions` — create an Action Taken.
- `GET /api/tickets/:ticketId/actions` — list Action Taken for a Ticket.
- `PATCH /api/tickets/:ticketId/actions/:actionId` — edit an Action Taken.
- `GET /api/reference/action-results` — list Result reference values.
- `PATCH /api/tickets/:ticketId/status` — general status transition (matrix-enforced); the Lab 3 `PATCH /api/tickets/:id/resolution` endpoint is kept and now also enforces BR-07.
- `GET /api/dashboard/requester`, `GET /api/dashboard/staff` (Administrator reuses the staff payload plus a user-count summary).

9. Acceptance Criteria

ID	Criterion
AC-01	Given a permitted IT Staff user and valid data, when an Actions Taken is created, then it is saved under the correct Ticket with the authenticated creator and approved assignee.
AC-02	Given an authenticated Requester, when dashboard data is retrieved, then only metrics and recent Tickets owned by that Requester are returned.
AC-03	Given Follow-Up Required is set to `true` with an empty Follow-up Note, when the record is submitted, then the request is rejected with a field-level `422` validation error.
AC-04	Given a Requester, when they call the create-Action-Taken API directly, then the request is rejected with `403 Forbidden` regardless of UI state.
AC-05	Given a Ticket with no `Resolved` -result Action Taken, when IT Staff attempts to transition it to `Resolved` , then the transition is rejected and the reason is shown.
AC-06	Given a Ticket with a `Resolved` -result Action Taken recorded by IT Staff, when IT Staff transitions it to `Resolved` , then the transition succeeds and the Ticket summary refreshes.
AC-07	Given two IT Staff editing the same Action Taken concurrently, when the second save uses a stale `updatedAt` , then the server returns `409 Conflict` and no data is silently overwritten.
AC-08	Given an authenticated IT Staff user, when the IT Staff Dashboard is requested, then Unassigned/My Assigned/status counts and recent Tickets reflect only Tickets that user may see, computed server-side.
AC-09	Given a Requester with zero Tickets, when the Requester Dashboard is requested, then every metric card renders its defined empty state without error.
AC-10	Given any dashboard metric card, when clicked, then the corresponding filtered Ticket list/queue opens showing the same record set implied by that metric.
AC-11	Given a Ticket with zero Action Taken records, when its Ticket Detail is viewed, then the Actions Taken area renders the defined empty state, not an error.
AC-12	Given a Cancelled or Closed Ticket, when viewed by IT Staff vs. Administrator, then only the Administrator sees a reopen control.

ID	Criterion
AC-13	Given a rapid double-click on "Save Action Taken," when two submissions race, then only one Action Taken record is created.

10. Definition of Done

- [ ] `specification.md`, `ui-spec.md`, `api-spec.md`, `tests.md` reviewed and merged to `lab4-staging` **before** any implementation PR is opened (tracked by Issue #53).
- [ ] All FR-01–FR-14 implemented and demonstrable.
- [ ] All BR-01–BR-16 enforced server-side (verified by tests, not just hidden UI controls).
- [ ] All AC-01–AC-13 covered by at least one automated test, traced in `tests.md`.
- [ ] Migration applied cleanly to a database seeded with Lab 1–3 data; rollback documented and tested.
- [ ] Idempotent seed script runs twice with no duplicate/erroring data.
- [ ] Zen Green visual/accessibility checklist completed for every new screen (see `ui-spec.md`).
- [ ] No console errors, dead links, placeholder text, or unfinished controls remain.
- [ ] README setup/seed/migration/test/demo instructions are current.
- [ ] `docs/lab-04/reviewer.md` and `docs/lab-04/ai-use.md` completed.

11. Assumptions and Decisions

- The Result of an Action Taken is a controlled reference value (§7.2), not free text, so the Resolved gate (BR-07) can match on it exactly.
- The Resolved gate ties directly to Actions Taken content — this is the specific "business rule spanning related records and state transitions" the handout's learning outcomes call for.
- Only Administrators can reopen a `Cancelled` / `Closed` Ticket; this is a deliberate exception to otherwise-symmetric IT Staff/Administrator permissions, matching the handout's note that Administrators "retain administrative access needed for support and testing."
- Both roles get an explicit "Dashboard" nav entry (the Lab 4 handout's Requester mockup shows "My Tickets" in the nav instead; treated as a simplification in the mockup image).
- Dashboards are new, standalone routes (`/dashboard`) added to the existing manual pathname-router (`window.history.pushState` / `popstate`), consistent with how Login, MyTickets, CreateTicket, and TicketDetail are already routed — no router library is introduced.
- The Actions Taken UI fills in the "Service Actions" tab on Ticket Detail that Lab 3 already added as a placeholder, rather than introducing a new tab.

- `recentTickets` on both dashboards is capped at 5 records per the handout's "concise, not entire collections" rule; "View all" drills into the existing Ticket Queue/My Tickets screen with the matching filter applied.